SureChEMBL 2.0
Now With Added Disease and Protein Annotations

Nicolas Bosc

Senior Scientific Developer

Cambridge Cheminformatics Meeting – 18/02/2026

EMBL-EBI

2nd ChEMBL User Group Meeting

We invite you to join the ChEMBL User Group Meeting (UGM), which is dedicated to building and supporting the ChEMBL and SureChEMBL user community.

June 10-11

EMBL-EBI

https://www.ebi.ac.uk/about/events/events/public-event/2026/ebi-9/

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Searching chemical patents is useful

- Freedom to operate
  - Identify blocking IP early
- Competitive intelligence
  - Monitor pipelines and emerging chemotypes
- State of the art
  - Often the earliest public disclosure
- Citations & references
  - Map technological lineage and impact
- Unique scientific knowledge
  - Novel compounds, scaffolds, reactions
  - Target–disease–indication relationships
    - ~3-year lead time over journals*

* Senger J Cheminform (2017) 9:26

Challenges of accessing patent data

- Documents are freely accessible (typically PDF or XML)
  - but largely unannotated
- Challenging documents
  - Often extremely long
  - Legal vocabulary ≠ scientific vocabulary
  - Information intentionally obfuscated
  - OCR artefacts and layout noise
- Hidden data
  - Chemical structures in images and tables
  - Inconsistent naming and representations between authorities
- Scale
  - >6 million patent documents published in 2025
  - Extensive patent families and duplication
  - Manual annotation at this scale is prohibitive

SureChEMBL in a nutshell

- Database of automatically annotated patents
- SureChem was developed by Digital Science, Macmillan Group
- Rebranded SureChEMBL, maintained and kept as it was by EMBL-EBI from 2013
- SureChEMBL2.0 introduced in 2025
- One of the last freely available source of annotated patents

122M
Patents

43M
Chemically-annotated documents

31M
Chemicals

1M
Biomedical entities

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What SureChEMBL does

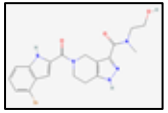
- Global patent coverage
  - USPTO, EPO, WIPO, CNIPA, JPO
- Automated full-text annotation
  - Chemical entities and biomedical terms extracted at scale
- Structure capture beyond text
  - Image-to-structure conversion
- Multiple searches
  - Keyword-based querying
  - Exact structure, substructure, similarity search
- Open and reusable data
  - CC BY 4.0 license
  - Full annotation releases every two weeks

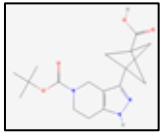
Compound identification

Fully automated pipeline

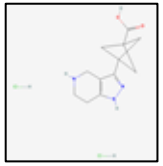
Patent data feed:
US, EU, CN, JP, WO
(IFI Claims)

image to structure





text to structure



Mol files
(US only)

RDKit ChEMBL

- Structure standardiser
- Registration hash code
- Property calculation

Chemical registration

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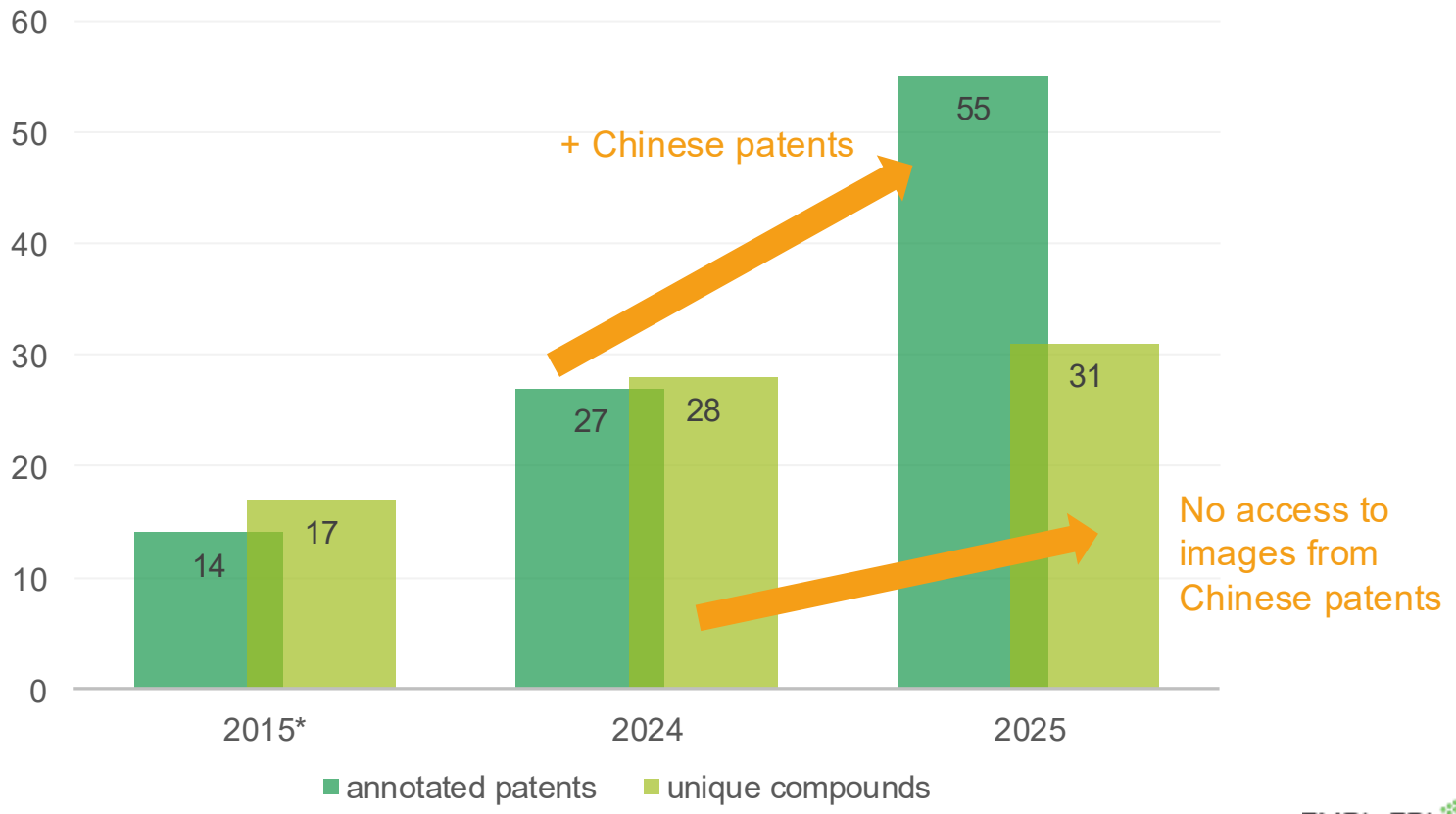
Data quantity evolution



*SureChEMBL paper - NAR, 2016, Vol. 44, D1220–D1228

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SureChEMBL2.0 — recent development

1. Annotation with Leadmine

2. Image2structure with MolScribe

3. Bulk data

SureChEMBL2.0

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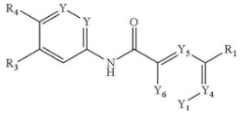
Compounds and compositions for inhibiting the activity of ABL1, ABL2 and BCR-ABL1

US-9458112-B2 / 2016-10-4

Abstract

The present invention relates to compounds of formula I:

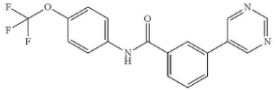
(I)



in which Y, Y_1, Y_4, Y_5, Y_6, R_1, R_2, R_3 and R_4 are defined in the Summary of the Invention; capable of inhibiting the activity of BCR-ABL1 and mutants thereof. The invention further provides a process for the preparation of compounds of the invention, pharmaceutical preparations comprising such compounds and methods of using such compounds in the treatment of cancers.

Claims

We claim:1. A compound, or a pharmaceutically acceptable salt thereof, selected from:



Legend

Chemical

Mode of A

Target

Disease

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Annotation with Leadmine

BEFORE

BIOMEDICAL

Genes/proteins, diseases, mechanism of actions

In house NLP model

No normalisation

CHEMICAL

Compound names, IUPAC names, SMILES, InChI

Chemaxon, Lexichem, OPSIN

Price, license conditions

NOW

BIOMEDICAL AND CHEMICAL

Genes/proteins, diseases, mechanism of actions, Compound names, IUPAC names, SMILES, InChI (+ others available)

*Leadmine**

Entity normalisation
Lower cost, favorable license

*Lowe and Sayle J Cheminf (2015), 7(Suppl 1):S5

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Annotation with Leadmine

Latest download!

- February release
  - Biomedical entity extraction across the full dataset through end of 2025
- Programmatic access available now
  - UI integration in progress
- Raw tool output provided
  - No post-processing or filtering

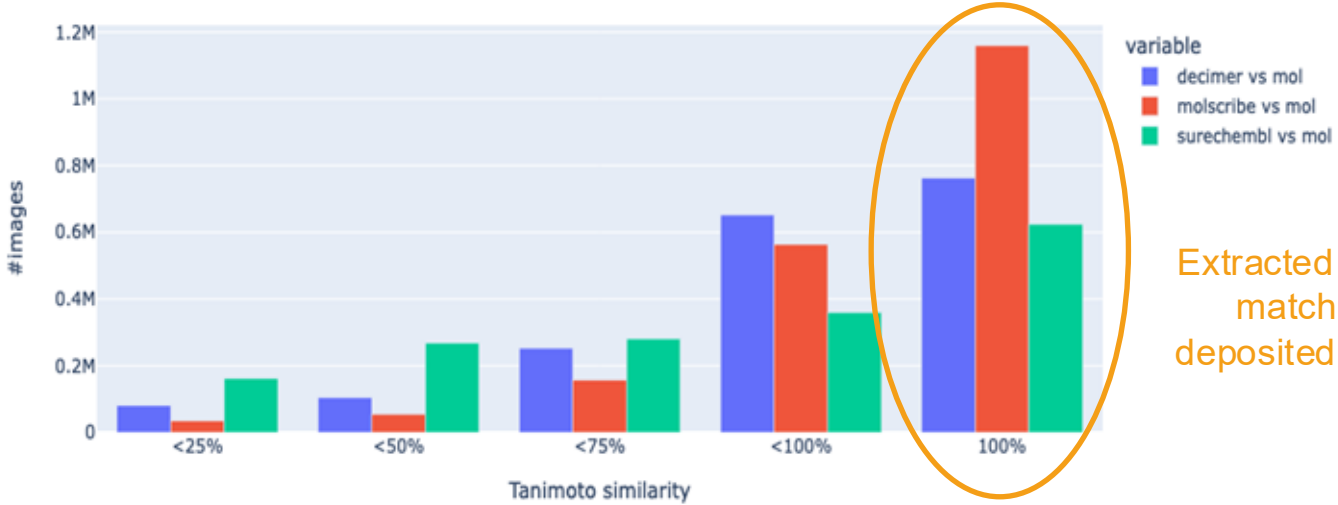
GeneOrProtein	913,773
Disease	137,749
Mechanism	8,202

patent_number	field_name	typename	corrected_text	resolved_form
"CN-105062961-A"	"clms"	"GeneOrProtein"	"SERPINA7"	"HGNC:11583"
"EP-4232089-A1"	"desc"	"GeneOrProtein"	"TNF-alpha"	"HGNC:11892"
"US-20030064926-A1"	"desc"	"GeneOrProtein"	"Proteinase K"	"P06873"
"US-20250179169-A1"	"desc"	"GeneOrProtein"	"tumor necrosis factor"	"HGNC:11892"
"EP-2166096-A2"	"desc"	"GeneOrProtein"	"T7 RNA polymerase"	"P00573"
"US-4201782-A"	"clms"	"Disease"	"nasopharyngitis"	"D009304"
"US-4094988-A"	"ttl"	"Disease"	"GASTRIC ULCERS"	"D013276"
"US-4753893-A"	"desc"	"Disease"	"INFLAMMATION"	"D007249"
"US-4170229-A"	"desc"	"Disease"	"SEBORRHEA"	"D012628"
"US-4690128-A"	"desc"	"Disease"	"sunburn"	"D013471"
"EP-0012316-A1"	"abst"	"Mechanism"	"SOLVENT"	""
"US-4225674-A"	"desc"	"Mechanism"	"ANTIBIOTIC"	""
"US-4148910-A"	"desc"	"Mechanism"	"BACTERICIDES"	""
"US-4268671-A"	"abst"	"Mechanism"	"DYES"	""
"US-4150655-A"	"abst"	"Mechanism"	"coolant"	""

Sample

Image compound extraction is being improved

- Explored published deep learning tools to improve quality of compound structures extracted from images
- All tested tools (including the ones currently used by SureChEMBL) return a majority of exact match
- MolScribe converts correctly significantly more images (and is faster)
- Finishing to integrate MolScribe in SureChEMBL pipeline

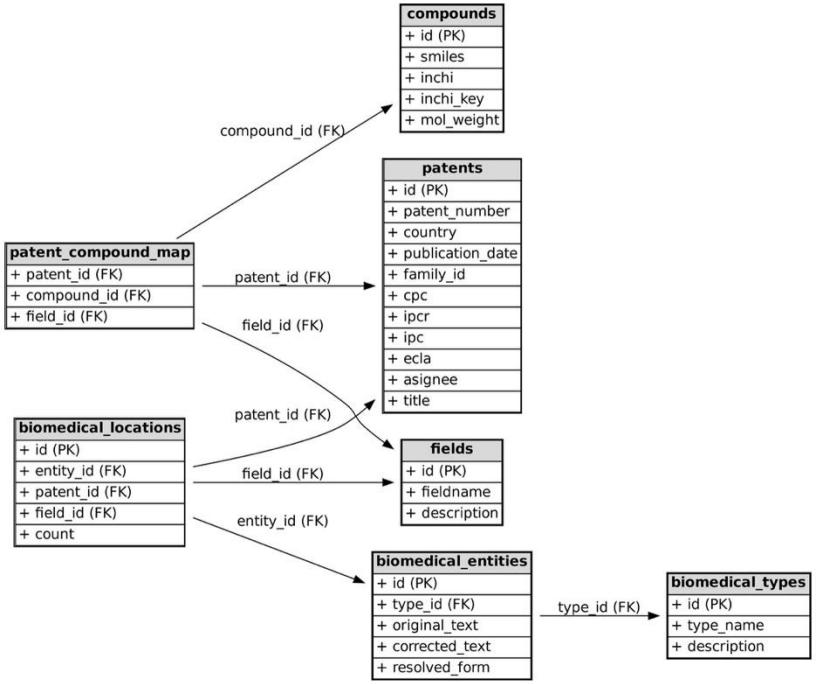


Extracted structures matching the deposited structures

EMBL-EBI

New bulk data download system

- Complete snapshot every 2 weeks
  - Core files representing the entire annotated SureChEMBL database
- Highly compressed, yet easily queryable
  - Parquet files designed for efficient large-scale analysis
- Structured data includes:
  - Patent metadata
  - Chemical annotations
  - Biomedical annotations
  - Annotation locations within patent text



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What questions can SureChEMBL answer?

- When was a given compound first reported?
- Which patents mention compound 1 and target A?
- Which diseases were reported first in 2025?
- Which proteins were reported first in 2025?

When was imatinib first reported?

```
"""
SELECT pat.*
FROM '{COMPOUNDS}' AS cpd
INNER JOIN '{PATENT_COMPOUND_MAP}' AS pcm ON cpd.id = pcm.compound_id
INNER JOIN '{PATENTS}' AS pat ON pcm.patent_id = pat.id
WHERE cpd.inchi_key = 'KTUFNOKKBVMGRW-UHFFFAOYSA-N'
ORDER BY publication_date ASC
LIMIT 1;
"""
```

id	patent_number	country	publication_date	family_id	cpc	ipcr	ipc	ecla	asignee	title
i64	str	str	date	i64	list[str]	list[str]	list[str]	list[str]	list[str]	str
8049323	"EP-0564409-A1"	"EP"	1993-10-06	4202064	["A61K 31/505", " A61P 35/00", ... " C07D 401/14"]	["A61K 31/505", " A61P 35/00", ... " C07D 403/04"]	["A61K 31/505", " C07D 401/04", ... " C07D 403/04"]	["C07D 401/04", " C07D 401/14"]	["CIBA-GEIGY AG (CH)", "CIBA GEIGY AG (CH)"]	"Pyrimidin derivatives and proc...

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How many patents mention imatinib with protein ABL?

```
"""
SELECT DISTINCT
    p.patent_number,
    p.publication_date,
    e1.corrected_text AS entity,
    c.inchi_key
FROM '{LOCATIONS}' AS l1
JOIN '{ENTITIES}' AS e1
    ON e1.id = l1.entity_id
JOIN '{PATENTS}' AS p
    ON p.id = l1.patent_id
JOIN '{PATENT_COMPOUND_MAP}' AS pcm
    ON pcm.patent_id = p.id
JOIN '{COMPOUNDS}' AS c
    ON c.id = pcm.compound_id
WHERE
    lower(e1.corrected_text) LIKE 'abl%'
    AND c.inchi_key = 'KTUFNOKKBVMGRW-UHFFFAOYSA-N'
ORDER BY publication_date
"""
```

shape: (21_936, 4)

patent_number	publication_date	entity	inchi_key
str	date	str	str
"EP-0564409-A1"	1993-10-06	"abl kinase"	"KTUFNOKKBVMGRW-UHFFFAOYSA-N"
"WO-1999003854-A1"	1999-01-28	"abl kinase"	"KTUFNOKKBVMGRW-UHFFFAOYSA-N"
"EP-0564409-B1"	2000-01-19	"abl kinase"	"KTUFNOKKBVMGRW-UHFFFAOYSA-N"
"EP-0998473-A1"	2000-05-10	"abl kinase"	"KTUFNOKKBVMGRW-UHFFFAOYSA-N"
"WO-2002022597-A1"	2002-03-21	"abl kinase"	"KTUFNOKKBVMGRW-UHFFFAOYSA-N"
…	…	…	…
"WO-2025262192-A1"	2025-12-26	"abl kinase"	"KTUFNOKKBVMGRW-UHFFFAOYSA-N"
"US-12509457-B2"	2025-12-30	"abl kinase"	"KTUFNOKKBVMGRW-UHFFFAOYSA-N"
"US-12509468-B2"	2025-12-30	"abl kinase"	"KTUFNOKKBVMGRW-UHFFFAOYSA-N"
"US-12509455-B2"	2025-12-30	"ABL1"	"KTUFNOKKBVMGRW-UHFFFAOYSA-N"
"US-12509455-B2"	2025-12-30	"ABL2"	"KTUFNOKKBVMGRW-UHFFFAOYSA-N"

How many patents mention imatinib in the claims with protein ABL in the title?

```
"""
SELECT DISTINCT
    p.patent_number,
    p.publication_date,
    p.title
FROM '{LOCATIONS}' AS l
JOIN '{ENTITIES}' AS e
    ON e.id = l.entity_id
JOIN '{PATENTS}' AS p
    ON p.id = l.patent_id
JOIN '{PATENT_COMPOUND_MAP}' AS pcm
    ON pcm.patent_id = p.id
JOIN '{COMPOUNDS}' AS c
    ON c.id = pcm.compound_id
JOIN '{FIELDS}' AS f1
    ON f1.id = l.field_id
JOIN '{FIELDS}' AS f2
    ON f2.id = pcm.field_id
WHERE
    lower(e.corrected_text) LIKE 'abl%'
    AND l.field_id = 4
    AND c.inchi_key = 'KTUFNOKKBVMGRW-UHFFFAOYSA-N'
    AND pcm.field_id = 1
ORDER BY publication_date
"""
```

shape: (234, 3)

patent_number	publication_date	title
str	date	str
"WO-2002102976-A2"	2002-12-27	"MUTATIONS IN THE BCR-ABL TYROS...
"WO-2003031608-A2"	2003-04-17	"MUTATED ABL KINASE DOMAINS"
"US-20030125305-A1"	2003-07-03	"Methods of treating leukemia"
"US-20030158105-A1"	2003-08-21	"Mutations in the Bcr-Abl tyros...
"EP-1436384-A2"	2004-07-14	"MUTATED ABL KINASE DOMAINS"
...	...	...
"CN-119242802-A"	2025-01-03	"System, kit and method for det...
"WO-2025049931-A1"	2025-03-06	"COMBINATIONS COMPRISING BCR:AB...
"US-20250149116-A1"	2025-05-08	"ABL1 FUSIONS AND USES THEREOF"
"CN-116536402-B"	2025-06-24	"Nucleic acid composition and k...
"WO-2025226724-A1"	2025-10-30	"BCR-ABL1 INHIBITORS FOR USE IN...

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Which diseases were first mentioned in 2025?

corrected_text	resolved_form	patent_count	patent_numbers	family_count	family_id
Camptodactyly taurinuria	C537972	1	WO-2025029943-A1	1	94395998
San Diego type skeletal dysplasia	C536670	1	US-12227500-B2	1	67805544
St. Helena familial genu valgum	C537685	1	WO-2025029943-A1	1	94395998
Velofacioskeletal syndrome	C536536	1	WO-2025132493-A1	1	94172898
cerebral intraventricular hemorrhage	D000074042	1	EP-4661893-A1	1	89853637

Which proteins were first mentioned in 2025?

corrected_text	resolved_form	patent_count	patent_numbers	family_count	family_id
Sso0001	Q97TX9	4	US-20250019744-A1, CN-119677865-A, CN-119948172-A, US-12359246-B2	2	88920123, 90275832
Sso1391	Q97YD4	4	US-20250019744-A1, CN-119677865-A, US-12359246-B2, CN-119948172-A	2	88920123, 90275832
HcCYP	Q4G338	3	CN-120005914-A, CN-115896136-B, CN-120005914-B	2	86488020, 95676120
bik4	S0DZI6	3	CN-119351477-A, CN-121087075-A, WO-2025059358-A1	3	94305184, 95021926, 97899895
Arlb protein	Q54UF1	2	CN-120000661-A, CN-120093737-A	2	95673297, 95877159

Unlocking early bioactivity data from patents

1 Untapped knowledge

Patents contain vast early-stage bioactivity data, often years before literature.

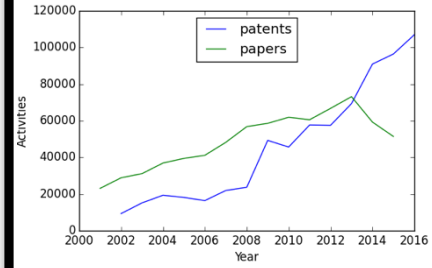


Fig 1. Amount of bioactivity data per (publication) year in ChEMBL versus extracted by LeadMine from US patents

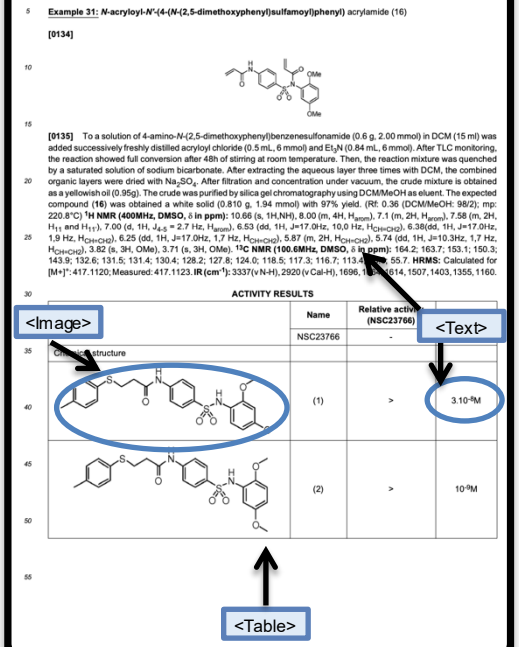
✓ Only 6% of compounds reported in patents were eventually found in scientific literature

✓ For 130 drug-target pairs, patents disclosed them on average 3.7 years earlier than scientific articles

Patent literature becomes a knowledge graveyard rather than a usable resource

2 Challenge

Patents are written for legal protection, not scientific clarity, leaving critical biological data buried in **unstructured and inconsistent text, tables, and images**.



Example 31: ***N*-acryloyl-*N'*-(4-(*N*-(2,5-dimethoxyphenyl)sulfamoyl)phenyl)** acrylamide (16)

[0134]

[0135] To a solution of 4-amino-*N*-(2,5-dimethoxyphenyl)benzenesulfonamide (0.6 g, 2.00 mmol) in DCM (15 ml) was added successively freshly distilled acryloyl chloride (0.5 mL, 6 mmol) and Et_3N (0.84 mL, 6 mmol). After TLC monitoring, the reaction showed full conversion after 48h of stirring at room temperature. Then, the reaction mixture was quenched by a saturated solution of sodium bicarbonate. After extracting the aqueous layer three times with DCM, the combined organic layers were dried with Na_2SO_4. After filtration and concentration under vacuum, the crude mixture is obtained as a yellowish oil (0.95g). The crude was purified by silica gel chromatography using DCM/MeOH as eluent. The expected compound **(16)** was obtained a white solid (0.810 g, 1.94 mmol) with 97% yield. (Rf: 0.36 (DCM/MeOH: 98/2); mp: 220.8°C) **^{1}H NMR (400MHz, DMSO,** δ **in ppm):** 10.66 (s, 1H,NH), 8.00 (m, 4H, H_{arom}), 7.1 (m, 2H, H_{arom}), 7.58 (m, 2H, H_{11} and $H_{11'}$), 7.00 (d, 1H, J_{4-5} = 2.7 Hz, H_{arom}), 6.53 (dd, 1H, J=17.0Hz, 10,0 Hz, $H_{CH=CH2}$), 6.38(dd, 1H, J=17.0Hz, 1,9 Hz, $H_{CH=CH2}$), 6.25 (dd, 1H, J=17.0Hz, 1,7 Hz, $H_{CH=CH2}$), 5.87 (m, 2H, $H_{CH=CH2}$), 5.74 (dd, 1H, J=10.3Hz, 1,7 Hz, $H_{CH=CH2}$), 3.82 (s, 3H, OMe), 3.71 (s, 3H, OMe). **^{13}C NMR (100.6MHz, DMSO,** δ **in ppm):** 164.2; 163.7; 153.1; 150.3; 143.9; 132.6; 131.5; 131.4; 130.4; 128.2; 127.8; 124.0; 118.5; 117.3; 116.7; 113.4; 56.4; 55.7. **HRMS:** Calculated for $[M+]^+$: 417.1120; Measured: 417.1123. **IR (cm^{-1}):** 3337(v N-H), 2920 (v Cal-H), 1696, 1[illegible]614, 1507, 1403, 1355, 1160.

ACTIVITY RESULTS

Compound Structure	Name	Relative activity (NSC23766)	
	NSC23766	-	
	(1)	>	3.10^{-9}M
	(2)	>	10^{-9}M

<Image>

<Text>

<Table>

3 Why we care

Bioactivity data is the foundation for:

- Drug design & optimisation
- SAR analysis
- Understanding mechanism of action in disease

4 Key Impacts

- Access data 3-10 years before scientific publications
- Enhance design-make-test-analyse (DMTA) cycles with richer starting knowledge
- Boost AI/ML models with real-world SAR data

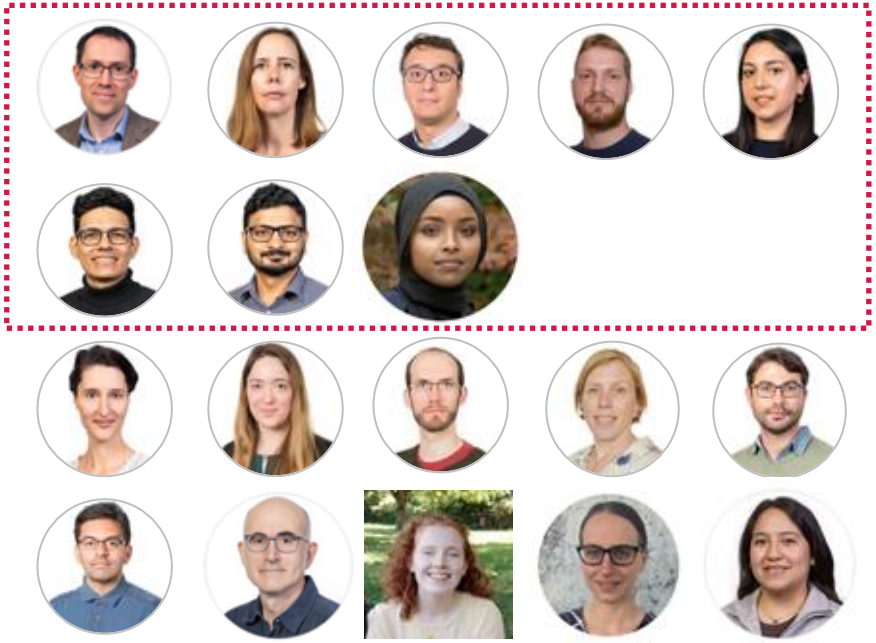
5 The Solution

To build an automated pipeline that can systematically extract, structure, and harmonise bioactivity data from patents.

SureChEMBL2.0 — recent developments

- Improved infrastructure
  - More stable, scalable and reliable production system
- Updated annotation workflow
  - Broader coverage, faster processing, more flexibility
- Bi-weekly full data release
  - Predictable access and reproducible analyses
- Next steps
  - Bioactivity extraction
  - UI refinements

Chemical Biology Services



EMBL

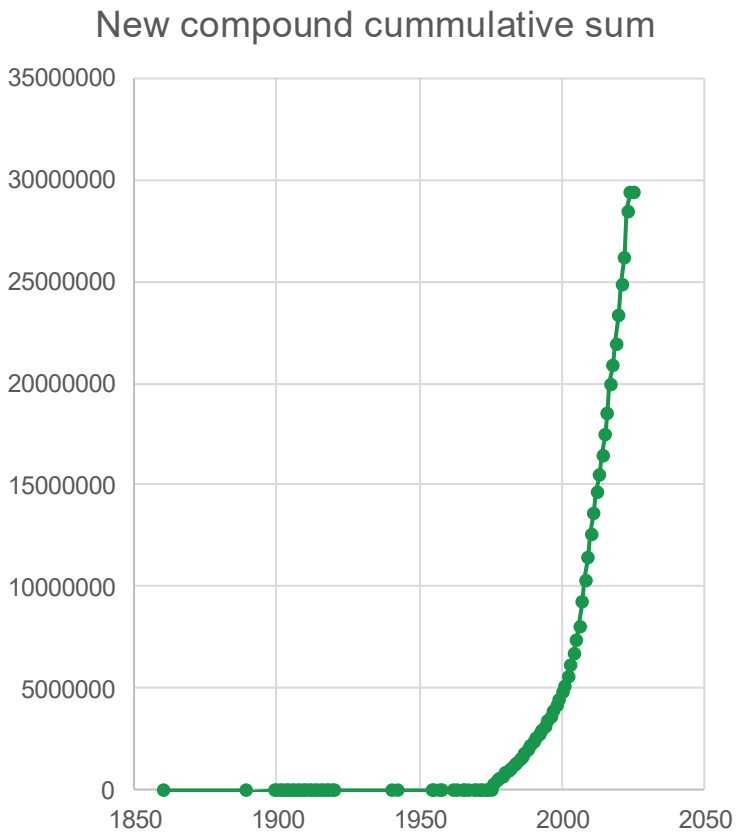
wellcome trust

Strategic Award

I will soon be leaving EMBL-EBI and am open to new positions
Feel free to connect

EMBL-EBI

Evolution of new compound identification



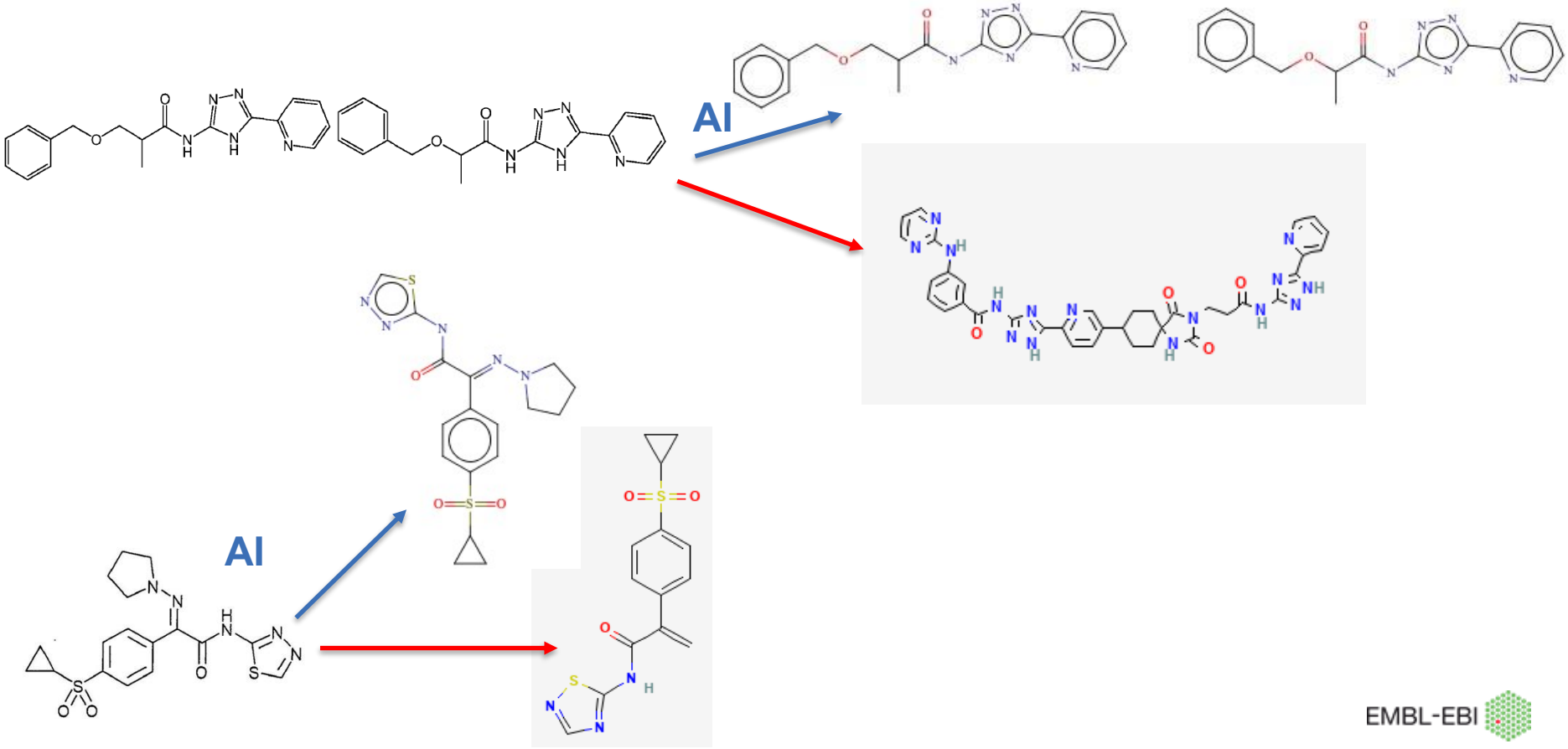
EMBL-EBI

Patent coverage

Authorities	Kind	Language	From	Full text	Biblio. data	Attachments	Annotated in SureChEMBL
CNIPA	*Applications*	*EN*	*1985*	*Yes (English translation)*	*Yes*	*No*	**2-7 days after receipt**
	Granted						
EPO	Applications	DE, EN, FR	1978	Yes	Yes	Yes	
	Granted		1980				
JPO	Applications	EN	1976	Yes (abstract)	Yes	No	
USPTO	Applications	EN	2001	Yes	Yes	Yes	
	Granted		1920-1949	Yes (abstract)	Yes	Yes (PDF)	
			1950-1975	Yes (abstract & claims)	Yes	Yes (PDF)	
			1976	Yes	Yes	Yes	
WIPO	Applications	EN, FR	1978	Yes	Yes	Yes	

EMBL-EBI

AI for Optical Chemical Structure Recognition



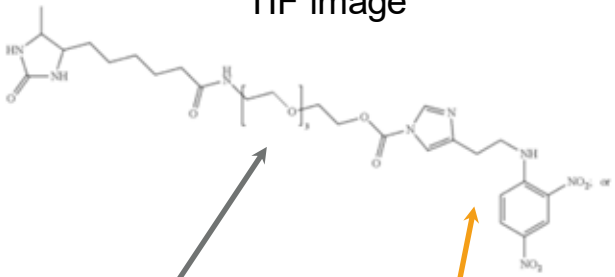
EMBL-EBI

Compound structure quality improvement

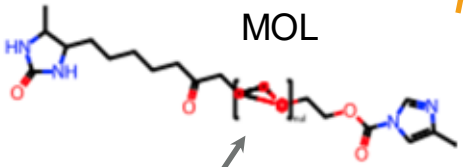
- Already good results but likely underestimate reality
- Structure in MOL file can be quite different from structure in the image…

PATENT

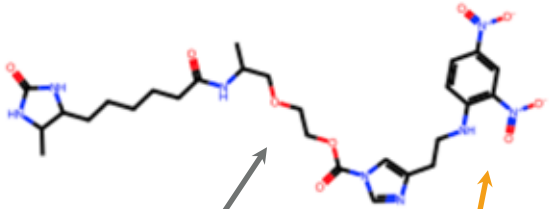
TIF image

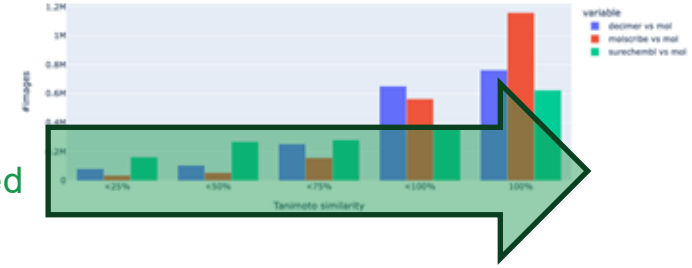


MOL



DECIMER





performance shift expected